

1. Customers Table

Column	Data Type	Constraints	Description
Customer_ID	INT	PK, AUTO_INCREMENT	Unique identifier for each customer
First_Name	VARCHAR(50)	NOT NULL	Customer's first name
Last_Name	VARCHAR(50)	NOT NULL	Customer's last name
Email	VARCHAR(100)	UNIQUE, NOT NULL	Customer's email address
Phone	VARCHAR(15)	UNIQUE, NOT NULL	Customer's contact number
Gender	ENUM('Male','Female','Other') NOT NULL		Gender of the customer
Date_of_Birth	DATE	NULL	Customer's date of birth
City	VARCHAR(50)	NOT NULL	City of residence
State	VARCHAR(50)	NOT NULL	State of residence
Registration_Date	DATE	DEFAULT (CURRENT_DATE)	Date the customer registered

2. Products Table

Column	Data Type	Constraints	Description
Product_ID	INT	PK, AUTO_INCREMENT	Unique identifier for each product
Product_Name	VARCHAR(100)	NOT NULL	Brand or marketing name of the product
SKU	VARCHAR(50)	UNIQUE, NOT NULL	Stock Keeping Unit code for inventory tracking
Category_ID	INT	FK (Category), NOT NULL	Links product to its main category grouping
Supplier_ID	INT	FK (Supplier), NOT NULL	Links product to its primary supplier
Cost_Price	DECIMAL(10,2)	NOT NULL, CHECK (Cost_Price > 0)	B2B purchase cost per unit
Selling_Price	DECIMAL(10,2)	NOT NULL, CHECK (Selling_Price > 0)	Retail shelf price before discounts

Current_Stock_Level	INT	NOT NULL, DEFAULT 0	Instant available unit count in warehouse
Reorder_Level	INT	NOT NULL, DEFAULT 10	Stock floor threshold triggering dynamic ordering
Status	ENUM('Active','Discontinued') DEFAULT 'Active'		Lifecycle status of the item

3. Categories Table

Column	Data Type	Constraints	Description
Category_ID	INT	PK, AUTO_INCREMENT	Unique identifier for each product category
Category_Name	VARCHAR(50)	UNIQUE, NOT NULL	Name of the business category
Descriptions	TEXT	NULL	Detailed breakdown of items within this category

4. Suppliers Table

Column	Data Type	Constraints	Description
Supplier_ID	INT	PK, AUTO_INCREMENT	Unique identifier for each supplier
Supplier_Name	VARCHAR(100)	NOT NULL	Official company name of the supplier
Contact_Name	VARCHAR(100)	NULL	Main point of contact name at supplier company
Email	VARCHAR(100)	UNIQUE, NOT NULL	Vendor communications email address
Phone	VARCHAR(15)	UNIQUE	Vendor contact phone number
City	VARCHAR(50)	NOT NULL	City where supplier warehouse is located
State	VARCHAR(50)	NOT NULL	State where supplier operates

5. Stores Table

Column	Data Type	Constraints	Description
Store_ID	INT	PK, AUTO_INCREMENT	Unique identifier for each physical location

Store_Name	VARCHAR(100) UNIQUE, NOT NULL	Branded name of the physical storefront
City	VARCHAR(50) NOT NULL	City where the store operates
State	VARCHAR(50) NOT NULL	Indian state location of the store
Opening_Date	DATE NOT NULL	Launch date of this physical retail branch

6. Employees Table

Column	Data Type	Constraints	Description
Employee_ID	INT	PK, AUTO_INCREMENT	Unique identifier for each employee
First_Name	VARCHAR(50)	NOT NULL	Employee's first name
Last_Name	VARCHAR(50)	NOT NULL	Employee's last name
Email	VARCHAR(100)	UNIQUE, NOT NULL	Employee's email address
Phone	VARCHAR(15)	UNIQUE, NOT NULL	Employee's contact number
Gender	ENUM('Male','Female','Other')	NOT NULL	Employee's gender
Role	VARCHAR(50)	NOT NULL	Workplace designation (e.g., Sales Associate, Cashier)
Salary	DECIMAL(10,2)	NOT NULL, CHECK (Salary > 0)	Fixed monthly base remuneration rate
Hire_Date	DATE	NOT NULL	Date employee joined the company workforce
Store_ID	INT	FK (Store), NOT NULL	Physical store branch assignment
Status	ENUM('Active','Inactive')	DEFAULT 'Active'	Work status indicating current employment

7. Orders Table

Column	Data Type	Constraints	Description
Order_ID	INT	PK, AUTO_INCREMENT	Unique identification code for the basket transaction

Order_Date	DATETIME	DEFAULT CURRENT_TIMESTAMP	Timestamp when checkout was logged
Customer_ID	INT	FK (Customer), NOT NULL	Customer who initiated the purchase
Store_ID	INT	FK (Store), NOT NULL	Retail branch processing this transaction
Employee_ID	INT	FK (Employee), NOT NULL	Cashier handling checkout point of sale
Gross_Amount	DECIMAL(10,2)	NOT NULL, CHECK (Gross_Amount >= 0)	Sum total of line prices before modifications
Discount_Amount	DECIMAL(10,2)	DEFAULT 0.00, CHECK (Discount_Amount >= 0)	Total retail price markdowns applied
Tax_Amount	DECIMAL(10,2)	NOT NULL, CHECK (Tax_Amount >= 0)	Aggregated GST calculations applied
Net_Amount	DECIMAL(10,2)	NOT NULL, CHECK (Net_Amount >= 0)	Final checkout billing amount paid
Order_Status	ENUM('Completed','Cancelled','Returned') DEFAULT 'Completed'		Global operational phase of the invoice transaction

8. Order_Items Table

Column	Data Type	Constraints	Description
Order_Item_ID	INT	PK, AUTO_INCREMENT	Unique structural line-item identifier

Order_ID	INT	FK (Orders), NOT NULL	The parent checkout transaction receipt number
Product_ID	INT	FK (Product), NOT NULL	The product SKU sold on this line
Quantity	INT	NOT NULL, CHECK (Quantity> 0)	Total volume units purchased of this product
Unit_Price	DECIMAL(10,2) NOT NULL		Individual retail price locked at purchase time
Total_Price	DECIMAL(10,2) NOT NULL		Quantity multiplied by actual unit price

9. Payments Table

Column	Data Type	Constraints	Description
Payment_ID	INT	PK, AUTO_INCREMENT	Unique reference identification code for the payment
Order_ID	INT	FK (Orders), UNIQUE, NOT NULL	Mandatory linked 1:1 order transaction invoice
Payment_Date	DATETIME	DEFAULT CURRENT_TIMESTAMP	Exact timeline stamp when transaction cleared
Amount_Paid	DECIMAL(10,2)	NOT NULL	Total financial tender received
Payment_Method	ENUM(('Cash','Credit Card','Debit Card','UPI'))	NOT NULL	Digital or cash clearing medium used
Payment_Status	ENUM('Success','Failed','Refunded')	DEFAULT 'Success'	Present settlement state of incoming ledger funds

10. Returns Table

Column	Data Type	Constraints	Description
Return_ID	INT	PK, AUTO_INCREMENT	Unique transaction tracking index for customer returns

Order_Item_ID	INT	FK (Order_Items), UNIQUE, NOT NULL	Mandatory linked 1:1 return-to-line-item index
Return_Date	DATETIME	DEFAULT CURRENT_TIMESTAMP	Clock timestamp when products were returned
Quantity_Returned	INT	NOT NULL, CHECK (Quantity_Returned > 0)	Units handed back by customer to branch
Refund_Amount	DECIMAL(10,2)	NOT NULL	Ledger cash returned back to client account
Reason_For_Return	VARCHAR(255)	NULL	Operational code or message for why return occurred